

DR. ANKUR AWADHIYA, I.F.S.

Deputy Director, Forest Survey of India

B. Tech (IIT Kanpur), Ph.D (IIT Kanpur)

AIGNFA (IGNFA Dehradun), PGDAWM (WII Dehradun)

DEDP (MIT), Prof. Cert. in Data Sci. & Comp. Sci. (Harvard)

Spec. in AI (Oxford), Mind and Decision-Making (Cambridge)

Adv. Digital Trans. (IIM A), Fin. Mgmt. (Duke), IP Law (U Penn)

I.F.S. Officer, Indian Forest Service 2014 batch, M.P. Cadre

Email: mp572@ifs.nic.in

Education and Training

Schooling

CBSE Board, D.A.V. Public School, Bilaspur, C.G.

- **Class Rank: 1**

- **CBSE merit certificates and laureate certificates** in classes 10th and 12th for top position in board exams.

- **KVPY fellowship** by the Dept. of Sci. and Tech., Govt. of India.

- **NTSE scholarship with 1st rank in the state.**

- Twice selected in the **Indian National Biology Olympiad.**

- **97% marks in the National Mathematics Olympiad.**

- **All-India Rank 3 in the National Science Olympiad.**

- **All-India Rank 2 and scholarship in All India Sanskrit Competitive exam** conducted by Arya Vidya Sabha, DAV - the largest nongovernmental educational body in India.

- **Invited to attend the Republic Day parade in the Prime Minister's box** due to my educational achievements.

2005-2009

B. Tech, Biological Sciences and Bioengineering, IIT Kanpur

- **Department Rank: 1**

- **Academic excellence award of IIT Kanpur** for best academic performance.

- **General Proficiency medal of IIT Kanpur** for best academic performance in the discipline.

- **Sri Binay Kumar Sinha Award of IIT Kanpur for the best B. Tech project across all departments** in the institute, for design and fabrication of novel, low-cost cryogel water-filtration devices utilising charged surfaces.

- **Proficiency medal of IIT Kanpur** for best project work in the department.

- **'Best Summer Innovation' award** at IIT Kanpur.

- Taught children from underprivileged sections of the society; the work received a **recommendation under the National Service Scheme.**

2009-2014

Ph.D, IIT Kanpur (Degree awarded at Convocation of 2015)

Thesis title: Studies in Agarose-based bioplastic material

- CPI: 10 / 10

- **Ranjan Kumar Memorial Award for the best socially relevant project in the graduating class**, for design and fabrication of new biodegradable plastics, anti-bacterial plastic surfaces, and drug-delivery vehicles for surgical dressings.
- **International patent** PCT/IB2015/053216 (02 May 2015)
- **Indian patent** 906/DEL/2015 (31 Mar 2015)

Sep 2014-Dec 2014	89th Foundation Course, Lal Bahadur Shastri National Academy of Administration, Mussoorie - Director's Gold Medal for securing the highest aggregate marks by the honourable President of India. - Director's Gold Medal in Indian History and Culture by the honourable President of India. - Director's Gold Medal in Public Administration by the honourable President of India. - First prize in Declamation Contest. - First prize in Photography Competition. - Second prize in Painting Competition. - Second prize in Movie Making Competition.
Dec 2014-May 2016	Associate of the Indira Gandhi National Forest Academy, Dehradun - All Round Outstanding Performance Gold Medal and prize for Outstanding achievement in Securing highest overall aggregate marks. - D. H. Kulkarni Gold Medal and prize for Outstanding achievement in Writing the Best Working Plan. - Sanjay Kumar Singh Memorial Gold Medal and prize for Outstanding achievement in Forest Management and Environmental Conservation. - Topper in Core Forestry Subjects Medal and prize for Outstanding achievement in Securing the highest marks in core forestry subjects. - Shri P. Srinivas Memorial prize for Outstanding achievement in Securing highest marks in Forest Health and Forest Law. - K. P. Sagriya Shreshta Vaniki prize for Outstanding achievement in All Academic Subjects. - Dr. B. N. Ganguli prize for Outstanding achievement in Academic Excellence. - R. N. Mathur Memorial prize for Outstanding achievement in Securing highest marks in Forest Management. - Senior Forester Prize for best field forester. - Associate of the Indira Gandhi National Forest Academy Honours diploma (equivalent of an M. Sc. in Forestry).
Sep 2017-Jun 2018	Post-Graduate Diploma in Advanced Wildlife Management, Wildlife Institute of India, Dehradun - Institute's Gold Medal for the Top trainee. - Book prize for the Top trainee in Wildlife Biology. - Honours diploma.

2021-2021	Professional Certificate in Data Science, Harvard University
2021-2022	Micromasters in Data, Economics, and Development Policy, Massachusetts Institute of Technology
2023-2024	Professional Certificate in Computer Science for Artificial Intelligence, Harvard University
2023-2024	Professional Certificate in Computer Science for Python Programming, Harvard University
2023-2024	Professional Certificate in Synthetic Aperture Radar, University of Alaska Fairbanks
2025-2026	Google AI Professional Certificate, Google

Specialisations (thematically arranged)

Forestry, Wetlands, and Natural Resources

- **Agroforestry** — University of Florida
- **Wetland Science and Conservation** — University of Illinois, Urbana–Champaign
- **Wetland Science and Conservation: Social Dimensions of Conservation** — University of Illinois, Urbana–Champaign
- **Wetland Science and Conservation: Fundamentals to Conservation** — University of Illinois, Urbana–Champaign

Management and Law

- **Financial Management** — Duke University
- **Alternative Investments** — University of Illinois, Urbana–Champaign
- **Advanced Digital Transformation** — IIM Ahmedabad
- **Conflict Management** — University of California, Irvine
- **Intellectual Property Law** — University of Pennsylvania

Computer Science, Neuroscience, and Artificial Intelligence

- **AI in Financial Services** — University of Oxford
- **Science of Mind and Decision Making** — University of Cambridge
- **Applied Machine Learning** — Johns Hopkins University
- **AI for Lawyers and Other Advocates** — University of Michigan
- **AI for Creative Work** — University of Michigan
- **AI Essentials** — Google

- **Data to Decision with AI** — Vanderbilt University
- **Python** — Arizona State University
- **Data Analysis with Python** — Google
- **Python Programming** — University of Colorado, Boulder
- **Scripting in Python** — Rice University

Remote Sensing and GIS

- **Geographic Information Systems (GIS)** — University of California, Davis
- **GIS, Mapping, and Spatial Analysis** — University of Toronto
- **Geospatial Techniques for Engineers** — L&T Edutech

Positions of responsibility

In the Indian Forest Service

- **Range incharge, Laugur (Territorial) Range, Balaghat:** May 2016 to December 2016.
 - Developed and deployed **SimplyFire**, a forest fire early detection and warning system that was later implemented throughout the state.
 - Developed and deployed a **simulation-based model of pine forest stands** to study the impacts of global warming on carbon sequestration and forest stand dynamics, including stand volume, mean annual increment, leaf area index, gross primary productivity and dry mass, later presented and published at the XIX Commonwealth Forestry Conference.
 - Developed and deployed **TripSafe**, a novel device to curb poaching of wild animals that was deployed in Tiger Corridors of Madhya Pradesh and countered electrocution-poaching.
 - Developed and deployed **AutoCam**, a camera-based surveillance and monitoring device, utilised for the study of natural behaviour of wild dogs (*Dholes*), and monitoring of entry points, cattle kills, water holes, salt licks and sensitive locations exceptionally susceptible to poaching activities.
 - As the **nodal officer for the conduction of medical camps under the Deen-dayal Vananchal Sewa Yojana**, played instrumental role in the conduction of medical camps in 30 forest villages in two divisions: North Balaghat (Territorial) Division and South Balaghat (Territorial) Division.
- **Sub-divisional Officer, Betul (Territorial) Sub-division, Betul and Officer Trainee, Wildlife Institute of India:** May 2017 to June 2017.
 - As a **member of the National Steering Committee for the Green Forestry sector** under the Global Technology Watch Group project supported by the Department of Science & Technology, Government of India, guided in scouting technology development happening globally and its prioritisation in the Indian context, used as an input to National Missions being implemented under the National Action Plan on Climate Change (NAPCC).

- Developed and deployed a web-based “**Madhya Pradesh Plant Biodiversity Search Engine**” to generate and query biodiversity data of the state, for simplification and consolidation of working plan preparation efforts and as an aid for *in-situ* and *ex-situ* conservation of vulnerable plant species of the state.
- **DCF, Green India Mission, Bhopal:** July 2018 to October 2018.
- **DFO, Nauradehi Wildlife Division, Sagar:** Oct 23 2018 to 17 June 2019.
 - Awarded the **Vanyaprani Sanrakshan Puraskar** by the Government of Madhya Pradesh for my contributions to the conservation of forest and wildlife.
 - Deployed several novel measures, including **DNA fingerprinting of trees, Drone and camera-based surveillance, Dog squad and legal tools including eco-sensitive zone notification** for the protection of Nauradehi Wildlife Sanctuary.
- **DCF IT and OSD to Hon’ble Forest Minister, Madhya Pradesh:** 17 June 2019 to 28 November 2023.
 - Awarded the prestigious **S. K. Seth prize in the field of Forestry** for my contributions on the **impact of climate change on wildlife health**.
 - Proposed and demarcated **11 Designer Wildlife Sanctuaries** in wildlife corridor areas of the state using GIS and high-resolution satellite data for inviolate area identification. **Dr. B. R. Ambedkar Sanctuary, Sagar** has been notified by the Government, and others are under process.
 - Developed and deployed **Nirnayan**, a 3-d decision-support system for site-suitability identification of projects for water conservation, protection, etc. in a landscape.
 - As a **nodal officer for forest fire management** in the state, aided in data-derived decision-making; wildfire alerts plummeted from 54,734 in the 2020–21 season to 34,559 in the 2021–22 season, and 16,647 in the 2022–23 season, and I was awarded the **Head of Forest Force’s Commendation** for my efforts.
- **Deputy Director, Forest Survey of India** from 29 November 2023 to date.
 - Developed and deployed **Anavaran**, an AI-based near-real-time deforestation alert system for the country utilizing multi-modal remote-sensing data and AI.

Academic

- **Instructor for Wildlife Conservation** online course, under Mission Karmayogi, Government of India
- **Instructor for Conservation Geography** online course, under Mission Karmayogi, Government of India
- **Instructor for Forests and their Management** online course, under Mission Karmayogi, Government of India
- **Instructor for Conservation Economics** online course, under Mission Karmayogi, Government of India
- **Instructor for Introductory Law For Conservation and Management** online course, under Swayam Prabha (broadcasted on DD Free Dish and Dish TV), Ministry of Education, Government of India

- **Instructor for Conservation Geography** online course, under Swayam Prabha (broadcasted on DD Free Dish and Dish TV), Ministry of Education, Government of India
 - **Instructor for Conservation Economics** online course, under Swayam Prabha (broadcasted on DD Free Dish and Dish TV), Ministry of Education, Government of India
 - **Instructor for Wildlife Ecology** online course, IIT Kanpur under the National Programme for Technology Enhanced Learning, Ministry of Human Resource Development, Government of India: Fall 2025, Fall 2024, Fall 2023, Fall 2022, Fall 2021, Fall 2020, Spring 2019
 - **Instructor for Conservation Geography** online course, IIT Kanpur under the National Programme for Technology Enhanced Learning, Ministry of Human Resource Development, Government of India: Fall 2025, Fall 2024, Fall 2023, Fall 2022, Fall 2021
 - **Instructor for Forests and their management** online course, IIT Kanpur under the National Programme for Technology Enhanced Learning, Ministry of Human Resource Development, Government of India: Spring 2025, Spring 2024, Spring 2023, Spring 2022, Spring 2021, Spring 2020
 - **Instructor for Conservation Economics** online course, IIT Kanpur under the National Programme for Technology Enhanced Learning, Ministry of Human Resource Development, Government of India: Spring 2025, Spring 2024, Spring 2023, Spring 2022, Spring 2021
 - **Instructor for Ecological Principles and Biodiversity for Sustainability** course, IIT Kanpur as Visiting Associate Professor of Practice: Spring 2025, Spring 2024; received **Director's commendation** for the same
 - **Instructor for Wildlife Conservation** online course, IIT Kanpur under the National Programme for Technology Enhanced Learning, Ministry of Human Resource Development, Government of India: Fall 2020, Fall 2019, Fall 2018
 - **Instructor for Forest Biometry** online course, IIT Kanpur under the National Programme for Technology Enhanced Learning, Ministry of Human Resource Development, Government of India: Spring 2020, Spring 2019, Fall 2017
 - **Coordinator of Seminar courses: BSE701 and BSE702** at IIT Kanpur for Ph.D students.
 - **Tutor for ESO219: Introduction to Biology** course at IIT Kanpur for B. Tech students: 2011-12, 2010-11
 - **Teaching assistant for BSE631: Developmental Biology II** course at IIT Kanpur for B. Tech, M. Tech and Ph.D students: 2010-11
 - **Teaching assistant for BSE292: Molecular Biology Lab** course at IIT Kanpur for B. Tech students: 2010-11, 2009-10
 - **Teaching assistant for ESO219: Introduction to Biology** course at IIT Kanpur for B. Tech students: 2009-10
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Patents and Publications

Patents

1. **906/DEL/2015** (31 Mar 2015): Indian patent
2. **PCT/IB2015/053216** (02 May 2015): International patent

Books

1. **Fundamentals of Machine Learning:** by Ankur Awadhiya and published by Taylor and Francis / CRC Press (2026)
2. **Wildfires: Principles, Management Strategies, and Best Practices:** by Ankur Awadhiya and published by Taylor and Francis / CRC Press (2025)
3. **Principles of Wildlife Conservation:** by Ankur Awadhiya and published by Taylor and Francis / CRC Press (2021)
4. **Fundamentals of distance sampling:** by Ankur Awadhiya and published by the Wildlife Institute of India, Dehradun (2018)
5. **QGIS based decision-support system for creating Forest Working Plans:** by Ankur Awadhiya and published by the Indira Gandhi National Forest Academy, Dehradun (2016)
6. **RS-GIS with free software:** by Ankur Awadhiya and published by the Indira Gandhi National Forest Academy, Dehradun (2016)
7. **Studies in Agarose-based bioplastic material (Ph.D thesis):** by Ankur Awadhiya (2014)

Departmental publications

1. **Firefighting — 10 lessons in management:** by Ankur Awadhiya and published by the Madhya Pradesh Forest Department (2023)
2. **Madhya Pradesh Forest Fire Compendium:** by Ankur Awadhiya and published by the Madhya Pradesh Forest Department (2022)
3. **Burn severity atlas of Madhya Pradesh 2022:** by Ankur Awadhiya and published by the Madhya Pradesh Forest Department (2022)
4. **Nightlight atlas of Madhya Pradesh:** by Ankur Awadhiya and published by the Madhya Pradesh Forest Department (2022)
5. **Height atlas of Madhya Pradesh:** by Ankur Awadhiya and published by the Madhya Pradesh Forest Department (2020)
6. **MNDWI atlas of Madhya Pradesh:** by Ankur Awadhiya and published by the Madhya Pradesh Forest Department (2020)
7. **Hillshade atlas of Madhya Pradesh:** by Ankur Awadhiya and published by the Madhya Pradesh Forest Department (2020)
8. **True colour composite atlas of Madhya Pradesh:** by Ankur Awadhiya and published by the Madhya Pradesh Forest Department (2020)

9. **NDVI atlas of Madhya Pradesh:** by Ankur Awadhiya and published by the Madhya Pradesh Forest Department (2020)
10. **Contour atlas of Madhya Pradesh:** by Ankur Awadhiya and published by the Madhya Pradesh Forest Department (2020)
11. **Transport atlas of Madhya Pradesh:** by Ankur Awadhiya and published by the Madhya Pradesh Forest Department (2020)
12. **Fire atlas of Madhya Pradesh:** by Ankur Awadhiya and published by the Madhya Pradesh Forest Department (2020)
13. **Deforestation in Madhya Pradesh: 2000-2018 An analysis of satellite-derived data:** by Ankur Awadhiya and Abhijit Awadhiya and published by the Madhya Pradesh Forest Department (2019)

Research articles

1. **Ankur Awadhiya**, Abhijit Awadhiya, Ajay Srivastava, Bilal Habib, Understanding Elephant Movement Paths in a Fragmented Corridor Using Animal Energetics, *Indian Forester*, 149(10), 1004-1009, 2023.
2. **Ankur Awadhiya**, Using energetics for Wildlife Population Management: Case study from Manas Tiger Reserve, *Proceedings of the International Conference on Wildlife Conservation*, 2023.
3. **Ankur Awadhiya**, 30 by 30 and Wildlife Population Management – Best practices from Madhya Pradesh, *Proceedings of the International Conference on Wildlife Conservation*, 2023.
4. **Ankur Awadhiya**, Use of novel devices for monitoring habitat use by wildlife and other ecological parameters – Case study from Madhya Pradesh, *Proceedings of the International Conference on Wildlife Conservation*, 2023.
5. **Ankur Awadhiya**, Understanding the differential roles of major causal factors in the 2017 Delhi smog through an analysis of satellite, meteorological and pollution data, *Indian Forester*, 145(8), 707-717, 2019.
6. **Ankur Awadhiya** & Ajay Srivastava, Rapid appraisal of Manas Tiger Reserve to understand its conservation challenges, *Indian Forester*, 144(10), 976-985, 2018.
7. **Ankur Awadhiya**, Impact of Climate Change on Wildlife Health: A manager's perspective, *Indian Forester*, 144(10), 911-921, 2018.
8. **Ankur Awadhiya**, Generation of division-level fire maps using satellite-derived fire data, *Indian Forester* 144 (5), 471-476, 2018.
9. **Ankur Awadhiya**, Impact of Global Warming on the Carbon Sequestration Potential and Stand Dynamics of Chir Pine Forests, *Indian Forester* 143(9), 907-914, 2017.
10. **Ankur Awadhiya**, Fire Breakers: Novel Devices for the Control of Forest Surface Fires, *Indian Forester* 143(5), 737-744, 2017.
11. **Ankur Awadhiya**, Adapting forest management to climate change, *National Seminar on Climate Change and Role of Communities for Adaptation and Mitigation* (2017)

12. **Ankur Awadhiya** & Abhijit Awadhiya, Impact of global warming on the carbon sequestration potential and stand dynamics of Chir Pine forests, *XIX Commonwealth Forestry Conference, 2017*
13. **Ankur Awadhiya**, Suhela Tyeb, Kalpana Rathore & Vivek Verma (2017), Agarose Bioplastic Based Drug Delivery System for Surgical and Wound Dressings. *Engineering in Life Sciences* DOI: 10.1002/elsc.201500116
14. **Ankur Awadhiya**, David Kumar, Kalpana Rathore, Bushara Fatma, & Vivek Verma (2016), Synthesis and characterization of agarose-bacterial cellulose biodegradable composites. *Polymer Bulletin* doi:10.1007/s00289-016-1872-3
15. **Awadhiya, Ankur**, Kumar, David & Verma, Vivek (2016), Crosslinking of agarose bioplastic using citric acid. *Carbohydrate Polymers* <http://dx.doi.org/10.1016/j.carbpol.2016.05.040>
16. Mohit Gera, **Ankur Awadhiya** & Neelu Gera (2016), Provenance trial of *Dalbergia sissoo* Roxb., *Indian Forester*, 142 (3) : 213-220, 2016
17. Mohit Gera, **Ankur Awadhiya** & Neelu Gera (2016), Progeny trial of *Acacia catechu*, *Indian Forester*, 142 (8) : 727-733, 2016
18. **Awadhiya, Ankur** & Verma, Vivek, Silver nano-particles incorporated agarose-based bioplastic films for packaging applications, *International Conference on Emerging Materials and Processes, 2014*
19. **Awadhiya, Ankur** & Verma, Vivek, A Novel Biocomposite Based On Agarose And Hydrolysed Cotton Cellulose For Packaging Applications, *International Conference on Emerging Materials and Applications, 2014*

General-interest articles

1. **Ankur Awadhiya**, Dharti ka itihaas – Bhoot tatha Samkaleen Mahaviluptiyaan (Hindi), *Madhya Pradesh Vananchal Sandesh, Oct-Dec 2022*
2. **Ankur Awadhiya**, Madhya Pradesh mein Van Agni Atlas ka srijan (Hindi), *Madhya Pradesh Vananchal Sandesh, Oct-Dec 2022*
3. **Ankur Awadhiya**, Madhya Pradesh mein burn severity aur uska atlas (Hindi), *Madhya Pradesh Vananchal Sandesh, Jul-Sep 2022*
4. **Ankur Awadhiya**, Madhya Pradesh Plant Biodiversity Search Engine (Hindi), *Madhya Pradesh Vananchal Sandesh, Jul-Sep 2022*
5. **Ankur Awadhiya**, Rekhiya avasanrachana evam paryavas vikhandan (Hindi), *Madhya Pradesh Vananchal Sandesh, Oct-Dec 2021*
6. **Ankur Awadhiya**, Nirnayan: 3d nakshon se van prabandhan ki madad (Hindi), *Madhya Pradesh Vananchal Sandesh, Oct-Dec 2021*
7. **Ankur Awadhiya**, Sansmaran: Tiffin (Hindi), *Aranya, 2020*
8. **Ankur Awadhiya**, Vyangya: Daastan-e-bureaucracy: Fursat Bahadur (Hindi), *Aranya, 2019*

9. **Ankur Awadhiya**, Vyangya: Daastan-e-bureaucracy: Hierarchy (Hindi), *Aranya*, 2019
 10. **Ankur Awadhiya**, Sansmaran: Ghonsala (Hindi), *Aranya*, 2018
 11. **Ankur Awadhiya**, Vichaar: Rescue (Hindi), *Aranya*, 2018
 12. **Ankur Awadhiya**, Himraj aur Shashank (Hindi), *Aranya*, 2018
 13. **Ankur Awadhiya**, Vyangya: Daastan-e-bureaucracy: Pita-Putra Prasanga (Hindi), *Aranya*, 2017
 14. **Ankur Awadhiya**, Vyangya: Daastan-e-bureaucracy: Nandi katha (Hindi), *Aranya*, 2017
 15. **Ankur Awadhiya**, Manzil abhi baaki hai Maa! (Hindi), *Aranya*, 2017
 16. **Ankur Awadhiya**, Vyangya: Hindi bhashi kshetron mein vyapt bhashayi avarodh (Hindi), *Aranya*, 2016
 17. **Ankur Awadhiya**, Vyangya: Daastan-e-bureaucracy: Kahani bandar ganana ki (Hindi), *Aranya*, 2016
 18. **Ankur Awadhiya**, Mera Panigrahan (Hindi), *Aranya*, 2015
 19. **Ankur Awadhiya**, Till we meet again, *Memoirs*, 2014
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Computer skills

I am comfortable on **Windows, Linux and Mac OSX platforms**, and have experience in the **following languages and packages**:

R, Python (Course work at Harvard Online)

Java (Course work at IIT Kanpur)

C, C++ (Certificate courses at IIT Kanpur)

Fluent, Gambit, MATLAB, python (Course work at IIT Kanpur)

HTML, PHP, My SQL (Certificate course at IIT Kanpur)

ArcGIS, QGIS and ERDAS packages, R programming (Course work at IGNFA, Dehradun, and Specialisations in GIS)

jemdoc and L^AT_EX

Bash scripting and server setups (I developed and deployed **backup servers, search engines, web servers and lab management system** during my graduate studies.)

Personal

Citizenship: Indian

Languages: Hindi (fluent), English (fluent), French (certificate course)

Marital status: Unmarried
